

Iyad Elwy

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Summary

I enjoy working on projects at the intersection of **data engineering**, **DevOps**, **GenAI**, and **infrastructure**. My interests include **container orchestration**, **GenAI workflow automation**, and building efficient, scalable software systems.

Employment History

Founding Software Engineer with a focus on modernization through **GenAI**
Index Labs – Reutlingen/München, Germany

- Built the complete CI/CD infrastructure using GitLab CI/CD, GitHub Actions, Docker, and Kubernetes across all company projects
- Led requirements engineering and system design, decomposing complex systems into scalable microservices architectures
- Developed production backend services with Python/FastAPI, PostgreSQL, Redis, and RabbitMQ for data processing
- Integrated enterprise systems including Oxaion ERP and Elasticsearch for automated data flows and process automation
- Leveraged AI-assisted development with Claude and advanced prompting techniques: Extended Thinking, Chain-of-Thought Reasoning
- Refactored monolithic systems into cloud-native architectures enabling flexible deployments across local, cloud, and Kubernetes environments
- Managed multi-environment deployments (Dev, Staging, Production) with configuration management and security practices
- Owned the full technical conception, implementation, and integration of the following AI projects — from requirements analysis and data preparation through LLM integration to production deployment:
 - Project KATA: Agentic AI pipeline that uses an LLM (AWS Bedrock) to process raw data from multiple sources in parallel and generate structured outputs
 - Project EIKE: GenAI service that automatically extracts order data from email attachments and forwards it to internal systems
 - Project LARA: Full-stack editorial platform (FastAPI + Next.js) with Elasticsearch integration and a multi-stage review workflow
 - Project DAVE: Internal service hub connecting heterogeneous enterprise systems (ERP, PIM, financial data) through a unified API layer

Oct 2025 – Current Date

Data Engineer for K8s & Intelligent (AI) Vehicle Interior Sensing Solutions
Bosch GmbH – Renningen, Germany

- Managed the lifecycle of Airflow deployments for data workflows
- Assisted with debugging Kubernetes deployments and resources alongside multinational teams
- Responsible for migrating Kubernetes resources between clusters across different versions
- Managed, renewed, and evaluated TLS certificates for deployments and services
- Introduced and configured Spark cluster mode for fast and efficient data processing and KPI computation
- Deployed Databricks Unity Catalog and migrated all tables for centralized, governance-compliant data management and access control
- Created and extended Helm charts for services and tools
- Wrote and debugged Python scripts for extracting, processing, and loading data into our data lake and data warehouse

Nov 2024 – Sep 2025

Data Science Internship
Bosch GmbH – Bühl, Germany

- Contributed to the development of predictive time-series algorithms to reduce the time-to-market of our product
- Implemented ETL pipelines for machine sensor data to establish a consistent and unified data state
- Implemented machine learning and deep learning models trained on machine sensor data
- Integrated Azure OpenAI with Microsoft Copilot to develop custom services that improved the team's daily productivity

Feb 2024 – Jun 2024 (5 Mo)

Education

M.Sc. Professional Software Engineering Oct 2025 – Sep 2027 Knowledge Foundation X Hochschule Reutlingen – Stuttgart, Germany Focus on Cloud-Native and scalable Software-Architecture	B.Sc. Informatics & Computer Science Apr 2020 – Jul 2024 German International University – Cairo, Egypt Major Data Science Minor Software Engineering	German International Abitur Apr 2010 – Jun 2019 Europa Schule Kairo – Cairo, Egypt Obtained my German International Abitur from Europa Schule Kairo
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Achievements & Publications

- Elwy, Iyad, & Sabty, Caroline. "Takween: A comprehensive Arabic collection and annotation tool." [Procedia Computer Science 244 \(2024\): 186-193.](#)

Professional Skills

Software: Python Java JavaScript SQL Node.js FastAPI NextJs Git Debian Based Linux Docker Airflow Github Actions Gitlab CI/CD PostgreSQL Redis RabbitMQ MongoDB RESTful APIs Rag	Data: Spark Pandas Polars Pytorch Scikit-Learn Data Preprocessing Data-Pipelining Visualization DuckDB Deep Neural Networks AWS Bedrock dbt Delta Lake Dagster ClickHouse Medallion Architecture	Infrastructure: Kubernetes Helm Terraform System Design Dev-Ops Workflows Cloud Design Patterns Docker Compose CI/CD Automation Multi-Environment Deployments Container Orchestration GitOps Azure CI/CD
AI & LLM: Prompt Engineering AI-Assisted Development Claude API	Leadership: Requirements Engineering Mentoring Agile/Scrum System Design	Deutsch (Native – Level C2) English (Fluent C1)